

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12407982

Kind Code

B1

Date of Patent

September 02, 2025

Inventor(s)

Thorson; Dean et al.

Optimal microphone selection on a remote speaker/microphone (RSM) assembly

Abstract

Techniques are disclosed for selecting one or more microphones on a remote speaker/microphone (RSM) assembly for input to an audio processing algorithm. An up orientation for the RSM is determined based on an embedded orientation sensor. For each microphone on the RSM, an up distance relative to the orientation sensor is determined based on a stored fixed mapping of microphone locations and current orientation of the orientation sensor, the microphone locations being fixed and known relative to the orientation sensor. Each microphone on the RSM is ranked from the uppermost to the lowermost. The uppermost microphone on the RSM assembly is then selected to serve as a voice input microphone while the microphone located furthest from the voice input microphone is selected to serve as a noise input microphone. A first audio input from the selected voice input microphone is then connected to an audio processing algorithm and a second audio input from the selected noise input microphone is connected to the audio processing algorithm.

Inventors: Thorson; Dean (Grayslake, IL), Koh; Bokchain (Crystal Lake, IL), Newton; Matthew (Park Ridge, IL), Oltmanns; Michael (Dyersville, IA)

Applicant: Relay, Inc. (Raleigh, NC)

Family ID: 96882207

Assignee: Relay, Inc. (Raleigh, NC)

Appl. No.: 19/233345

Filed: June 10, 2025

Publication Classification

Int. Cl.: H04R3/00 (20060101)

U.S. Cl.:

CPC **H04R3/005** (20130101);

Field of Classification Search

CPC: H04R (3/005)

USPC: 381/56; 381/58; 381/91; 381/94.1; 381/94.7

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0190769	12/2008	Wang	381/66	H03G 3/32
2013/0216050	12/2012	Chen	381/56	H04B 3/23
2013/0272540	12/2012	Hgren	381/94.1	H04R 3/00
2013/0332156	12/2012	Tackin	704/226	H03G 3/3005
2015/0172811	12/2014	Sassi	381/71.1	H04M 1/6041
2019/0069083	12/2018	Salehin	N/A	H04R 3/005
2021/0144469	12/2020	Zhao	N/A	H04R 1/1075

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2583028	12/2019	GB	H04S 3/02

Primary Examiner: Jerez Lora; William A

Background/Summary

BACKGROUND OF THE INVENTION

(1) Portable communication devices (PCDs) like the ones made by Relay, Inc. operate like walkie-talkies. A user presses and holds a button to transmit and releases to receive audio. The user typically will hold the PCD up in proximity but not too close to the mouth for optimal audio quality. Often, however, a user may wish to affix or clip the microphone and speaker to clothing on their upper torso making for a more convenient mode of operation. Doing so necessitates the use of an accessory coupled with the PCD termed herein as a remote shoulder microphone (RSM). It is impractical to affix the PCD to the clothing in that particular area of the torso.

(2) As mentioned, the RSM is normally mounted on the upper torso of a user's clothing or uniform, placing the microphone(s) of the RSM below but still near the user's mouth. However, because of personal preference, the orientation of the RSM can vary. While the RSM may have a standard orientation (i.e., up/down/left/right), the actual orientation may not align depending on how the user chooses to wear the RSM. For instance, the RSM typically includes a cord/cable connecting the RSM with the PCD. The PCD may be more typically affixed at belt level. In one configuration, the RSM may then be worn over the shoulder with the cable running up the back of the user. This inverts the up/down/left/right orientation of the RSM. Sometimes the RSM may be affixed in an askew (e.g., sideways) position which may also materially differ from the standard orientation.

(3) The RSM may be equipped with echo canceling and noise reduction algorithms designed to

enhance the audio quality of the RSM. Echo cancellation primarily tackles the issue of sound from a speaker being picked up by a microphone and retransmitted, causing a distracting echo. Algorithms for this typically involve an adaptive filter that models the echo path and generates an estimate of the echo. This estimated echo is then subtracted from the microphone's input signal. Noise reduction focuses on reducing unwanted background sounds to make desired audio like speech more intelligible. Common noise reduction methods include spectral subtraction, where an estimate of the noise's frequency spectrum is removed from the overall audio spectrum. Other approaches involve various types of filters, including adaptive filters, and increasingly, machine learning algorithms.

(4) Lots of microphone assemblies (including the RSM described herein) utilize audio processing algorithms such as, for instance, echo cancellation and noise suppression (ECNS) to improve the sound quality of the audio that is input. The audio processing algorithms generally require two microphones to function. A first microphone is for voice input and a second microphone is for noise input. The microphones are normally physically mounted opposite one another on the device for acoustic isolation.

(5) Sometimes, like in the inverted position described earlier, the fixed voice input microphone can be positioned away from the mouth, with the noise microphone positioned close. This positioning confuses the audio processing software and minimizes its effectiveness. The solutions described and claimed herein are directed toward the selection of the voice microphone vs. the noise reduction microphone when implementing an audio processing algorithm such as ECNS—not the actual ECNS algorithms. The specification describes a technique using a gyroscope and accelerometer sensor and measured phone orientation to choose the input and noise reduction microphones, not digital signal processing.

(6) Other techniques utilize a hardware based dual-microphone digital signal processing selection algorithm that estimates speech energy on both microphones before selecting which one to use as the input microphone and which one to use as the noise suppression microphone. This technique is typically performed relatively slowly in an attempt to minimize power consumption.

(7) What is needed is a means for selecting, on an RSM device, which microphone should be the input microphone and which microphone should be the noise suppression microphone for a noise reduction algorithm such as ECNS that can switch among a plurality of microphones faster and with much less power used.

SUMMARY OF THE INVENTION

(8) The embodiments herein describe a method, system, and computer program product for selecting one or more microphones on a remote speaker/microphone (RSM) assembly for input to a noise reduction algorithm.

(9) In one embodiment, a method comprises determining an up orientation for the RSM based on an embedded orientation sensor. This is followed by accessing a stored fixed mapping of microphone locations, the microphone locations being fixed and known relative to the orientation sensor. For each microphone on the RSM, an up distance may be determined relative to the orientation sensor based on the stored fixed mapping of microphone locations and current orientation of the orientation sensor. Each microphone on the RSM may be ranked from the uppermost to the lowermost. The uppermost microphone on the RSM assembly may then be selected to serve as a voice input microphone. The microphone located furthest from the voice input microphone may be selected to serve as a noise input microphone. A first audio input from the selected voice input microphone may be connected to a noise reduction algorithm. A second audio input from the selected noise input microphone may be connected to the noise reduction algorithm.

(10) In another embodiment, the orientation sensor includes at least one of an accelerometer or a gyroscope.

(11) In yet another embodiment, the noise reduction algorithm may be one of an echo cancellation

and noise suppression (ECNS) algorithm or a speech recognition algorithm.

(12) In still another embodiment, the RSM assembly includes an even number of microphones organized in pairs, wherein each microphone pair is fixed opposite one another where the first microphone in the pair is furthest from the second microphone in the pair such that any other microphones on the RSM assembly are closer to the either microphone in the pair.

(13) In still another embodiment, the internal mapping of microphone locations and orientation sensor is a 3-dimensional database.

(14) In still another embodiment, the steps of the method are repeated upon the detection of motion based on output from the orientation sensor.

(15) In yet another embodiment, a system comprised of a set of one or more processors; and a memory operatively coupled to the set of one or more processors and comprising code executable by the set of one or more processors to determine an up orientation for the RSM based on an embedded orientation sensor. The system may access a stored fixed mapping of microphone locations, the microphone locations being fixed and known relative to the orientation sensor. For each microphone on the RSM, the system may determine an up distance relative to the orientation sensor based on the stored fixed mapping of microphone locations and current orientation of the orientation sensor. The system may rank each microphone on the RSM from the uppermost to the lowermost. The system may then select the uppermost microphone on the RSM assembly to serve as a voice input microphone. The microphone located furthest from the voice input microphone may be selected by the system to serve as a noise input microphone. A first audio input from the selected voice input microphone may be connected to a noise reduction algorithm and a second audio input from the selected noise input microphone may be connected to the noise reduction algorithm by the system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A illustrates an exterior top view of an RSM assembly according to an embodiment of the invention.

(2) FIG. 1B illustrates an exterior side view of an RSM assembly according to an embodiment of the invention

(3) FIG. 1C illustrates some of the interior components of an RSM assembly according to an embodiment of the invention

(4) FIG. 2A illustrates a top view of the inside of an RSM assembly showing microphone and accelerometer placement according to an embodiment of the invention.

(5) FIG. 2B illustrates a first side view of the inside of an RSM assembly showing microphone and accelerometer placement according to an embodiment of the invention.

(6) FIG. 2C illustrates a second side view of the inside of an RSM assembly showing microphone and accelerometer placement according to an embodiment of the invention.

(7) FIG. 3 illustrates the accelerometer in four different exemplary orientations according to an embodiment of the invention.

(8) FIG. 4 illustrates a perspective view of the inside of an RSM assembly showing microphone and accelerometer placement with respect to an x-y-z coordinate system according to an embodiment of the invention.

(9) FIG. 5 illustrates a user wearing the RSM in one exemplary orientation according to an embodiment of the invention.

(10) FIG. 6 illustrates a logic flow diagram describing the selection of a microphone based on the orientation of the RSM with a user according to an embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(11) The present invention relates to determining which microphone should be the input microphone and which microphone should be the noise suppression microphone for an ECNS algorithm to optimize noise canceling for a remote shoulder microphone (RSM) assembly. The RSM is coupled (wired or wirelessly) to a personal communication device (PCD) to allow for more convenient usage of the PCD for some users.

(12) There are two approaches considered herein to enhance the effectiveness of the noise reduction algorithms for devices like the RSM assembly. A first approach would be to perform the noise reduction algorithm twice for each normal optimization period, then choose the input based upon the results (i.e., the solution with the best noise suppression). This approach requires relatively significant digital processing power, materially impacts battery life, and is not guaranteed to provide an optimum result.

(13) The second approach, which is the subject of this disclosure, is to eliminate the complex digital signal processing and use the physical characteristics of the device and its internal sensors to choose the proper microphone to be the “voice” input and the proper microphone to be the noise suppression input. Because the positioning of the RSM on the body of the user will always be below the mouth level, a physical orientation determination solution is possible.

(14) The RSM assembly described herein includes multiple (even numbered) microphones as well as a sensor such as an accelerometer/gyroscope. Each microphone pair is situated on the device opposite of one another. A three-dimensional orientation among the sensor and the plurality of microphones is fixed and known and may be stored in a microphone location database.

(15) Selecting the microphone to use as voice input involves determining, via the sensor, which direction is up. Once that is known, a lookup into the microphone location database is made that determines which microphone is currently the furthest in the “up” position based on the fixed and known orientation. This microphone is then deemed the best for voice input. The noise suppression microphone is then chosen as the one opposite to (i.e. furthest from) the voice input microphone. This is a fast and extremely low-power solution that results in the optimum selection for voice input.

(16) FIG. 1A illustrates an exterior top view of an RSM **100** according to an embodiment of the invention. In this view, the RSM **100** includes four (4) microphones **105a-d**, a pattern of holes **110** to allow for a speaker assembly (not shown) to output audio, a push-to-talk (PTT) button **120**, and a hidden port **130**. The hidden port **130** may be used to power/recharge the RSM **100** as well as couple the RSM **100** to the PCD (not shown) via a cable (not shown). The PTT button **120** is used by the user to engage and disengage transmission mode for the PCD via the RSM **100**. The RSM **100** further includes one or more processors for executing one or more algorithms, including those mentioned in this description.

(17) FIG. 1B illustrates an exterior side view of the RSM **100** according to an embodiment of the invention showing another view of PTT button **120** and port **130**.

(18) FIG. 1C illustrates some of the interior components of an RSM **100** according to an embodiment of the invention. In addition to the previously identified microphone(s) **105a-d**, the RSM **100** further includes one or more speakers **155**, one or more processors **150**, one or more sensors **160**, a memory component **165**. The memory component may store mapping data **170** and one or more noise reduction algorithms **175**.

(19) The mapping data **170** may comprise data points that indicative of a three-dimensional orientation of the microphones **105a-d** relative to the sensor(s) **140**. Mapping data **170** can be used to determine which particular microphone among **105a-d** is positioned most upward given a current sensor **140** orientation. Noise reduction algorithms **175** may comprise, for instance, echo cancellation and noise suppression (ECNS) algorithms that receive voice input from a first microphone and noise input from a second microphone. The selection of which microphone to serve as the voice input and which microphone to serve as the noise input is the subject of the present description. The sensor(s) **140** may include either or both of a gyroscope or an

accelerometer. In operation, processor(s) **150** receive orientation data from sensor(s) **140** to determine which direction is currently up. Processor(s) **150** may then consult mapping data **170** to determine which microphone **105a-d** is currently the most upward based on the determined current sensor **140** orientation. This microphone is then deemed the voice input while its paired microphone is deemed the noise input microphone by virtue of being located furthest from the voice input microphone. Processor(s) **150** then process, using the noise reduction algorithms **175**, the input signal from the voice input and noise input microphones.

(20) FIG. 2A illustrates a top view of the inside of an RSM **100** showing microphone **105(a-d)** and accelerometer **140** placement according to an embodiment of the invention.

(21) FIG. 2B illustrates a first side view of the inside of an RSM **100** showing microphones **105(a-d)** and sensor **140** placement according to an embodiment of the invention. The sensor **140** may be an accelerometer and/or a gyroscope and is affixed in a known position and orientation relative to microphones **105a-d**.

(22) FIG. 2C illustrates a second side view of the inside of an RSM **100** showing two microphones (**105d** and **105b**) and sensor **140** placement according to an embodiment of the invention. In this depiction, a first vertical distance (**d1**) upward between sensor **140** and microphone **105d** and a second vertical distance (**d2**) downward between sensor **140** and microphone **105b** is illustrated.

(23) FIG. 3 illustrates a sensor **140** in four different exemplary orientations within the RSM **100** according to an embodiment of the invention. Each exemplary sensor orientation includes an X-Y-Z coordinate axis in which the z-axis is labeled as “up”. In the first exemplary orientation (a), sensor **140** may be characterized as in a standard upright orientation based on the front/back characterization of the RSM **100** which would correspond to the RSM **100** being mostly upright. In the second exemplary orientation (b), sensor **140** may be characterized as tilted slightly to the right or slightly downward based on the front/back characterization of the RSM **100** which would correspond to the RSM **100** being angled as shown. In the third exemplary orientation (c), sensor **140** may be characterized as tilted more to the right or more downward based on the front/back characterization of the RSM **100** which would correspond to the RSM **100** being angled as shown. In the fourth exemplary orientation (d), sensor **140** may be characterized as tilted to the left or downward based on the front/back characterization of the RSM **100** which would correspond to the RSM **100** being angled as shown.

(24) FIG. 4 illustrates a perspective view of the inside of an RSM **100** showing microphone **105(a-d)** and sensor **140** placement with respect to an x-y-x coordinate system according to an embodiment of the invention. In this depiction, a first vertical distance (**d1**) upward between sensor **140** and microphone **105a**, a second vertical distance (**d2**) upward between sensor **140** and microphone **105c**, a third vertical distance (**d3**) downward between sensor **140** and microphone **105d**, and a fourth vertical distance (**d4**) downward between sensor **140** and microphone **105b** is illustrated. Based on the orientation of the RSM **100** relative to the X-Y-UP axis, microphone **105c** is the furthest “up” and would be the microphone selected as the voice input microphone for the RSM **100**. Microphone **105a** is paired with microphone **105b** meaning they are separated the furthest from one another. Similarly, microphone **105c** is paired with microphone **105d** such that they too are separated the furthest from one another. Thus, whichever microphone **105a-d** is selected as the voice input by virtue of being the most “up”, its paired microphone will be selected as the noise suppression microphone for purposes of ECNS algorithms.

(25) FIG. 5 illustrates a user wearing the RSM **100** in one exemplary orientation according to an embodiment of the invention. In this example, the user has chosen to wear the RSM **100** close to a standard upright orientation rather than an inverted over the shoulder orientation. The four microphones (**105a-d**) are shown simply as a, b, c, and d for simplicity. Microphone (a) is in the most upright position followed by (c), (d), and (b). Thus, microphone (a) will be selected as the voice input microphone. Moreover, the microphones have been paired together—(a) with (b) and (c) with (d)—such that they are physically located opposite and furthest away from one another.

Therefore, when microphone (a) is chosen as the voice input, microphone (b) will automatically be selected as the noise suppression input microphone for the ECNS algorithm(s).

(26) FIG. 6 illustrates a logic flow diagram describing the selection of a microphone **105(a-d)** based on the orientation of the RSM **100** with a user according to an embodiment of the invention. More specifically, the logic flow diagram describes the selection of a voice input microphone and a noise suppression input microphone to be used as the input microphones for echo cancellation and noise suppression (ECNS) algorithm(s).

(27) In block **605**, the RSM **100** has stored within local memory a mapping of the microphone locations **105a-d** and the sensor **140** within the RSM **100**. This mapping data **170** may be embedded in memory **165** based on a known fixed orientation among the microphones **105a-d** and the sensor(s) **140**.

(28) In block **610**, the “up” orientation of the RSM **100** may be determined based on the current sensor **140** orientation. Once the up orientation of sensor **140** is determined, processors **150** may determine which microphone among **105a-d** is positioned uppermost based on mapping data **170** in block **615**.

(29) If this is the initial instance such as the beginning of a conversation using the RSM **100**, the process may skip ahead to block **630**. However, if a pair of microphones (voice and noise) has already been selected, a determination may be made, in decision block **620**, as to whether the microphone currently in the upmost position as determined in block **615** is different from the microphone that is currently being used as the voice input microphone. If they are the same, control of the process forks to block **625** where it waits until the sensor **140** detects motion that may be indicative of a change in orientation. Once such motion is detected, control returns to block **610** to determine the up orientation for sensor **140**.

(30) If the microphone in decision block **620** is different from the current voice microphone, the microphone determined in block **615** may then be selected as the new voice input microphone in block **630**. Selecting this microphone as the most optimal will always be true because the RSM **100** will always be positioned beneath the user's mouth. Thus, the uppermost microphone is, by default, the one closest to the user's mouth meaning it is most appropriate to serve as the voice input microphone. In addition, the microphone opposite to the voice input microphone is the ideal microphone to serve as the noise input microphone due to its fixed location furthest from the selected voice input microphone. The opposite microphone is then selected as the noise input microphone in block **635**. Recall, microphones **105a-d** have been arranged in pairs **105a** with **105b** and **105c** with **105d**. These pairs have been specifically linked as they are fixed opposite and furthest away from one another such that in any orientation microphone **105b** will always be further from microphone **105a** than microphones **105c** and **105d**. Similarly, microphone **105c** will always be further from microphone **105d** than microphones **105a** and **105b**.

(31) Once the voice input and noise input microphones have been determined, each is connected via the processor(s) **150** to the noise reduction algorithms **175** in block **640** using a dynamic audio routing process. Dynamic audio routing propagates the signal from the selected microphone (e.g., **105a-d**) to the processor(s) **150**. Dynamic audio routing captures a raw audio stream (e.g., voice and noise) at the respective selected microphones and directs the raw audio to a noise reduction algorithm. The noise reduction algorithms may then process the raw audio inputs. The processed audio outputs may then be routed to their final destination to provide the best results regarding audio clarity and noise cancellation for the RSM **100**.

(32) One such noise reduction algorithm may be an echo cancellation and noise suppression (ECNS) algorithm. Another such noise reduction algorithm may be a speech recognition algorithm.

(33) In the claims, any reference signs placed between parentheses shall not be construed as limiting the claim. The word “comprising” or “including” does not exclude the presence of elements or steps other than those listed in a claim. In a device claim enumerating several means, several of these means may be embodied by one and the same item of hardware. The word “a” or

“an” or “the” preceding an element does not exclude the presence of a plurality of such elements. The mere fact that certain elements are recited in mutually different dependent claims does not indicate that these elements cannot be used in combination. The word “about” or similar relative term as applied to numbers includes ordinary (conventional) rounding of the number with a fixed base such as 5 or 10.

(34) It is worth noting that while specific blocks are used in the figures, and a particular ordering of blocks has been illustrated, these are non-limiting examples. In certain contexts, two or more blocks may be combined, a block may be split into two or more blocks, or certain blocks may be re-ordered or re-organized or omitted as appropriate, as the explicit illustrated examples are used only for descriptive purposes and are not to be construed as limiting.

(35) As used herein, the statement that two or more parts or components are “coupled” shall mean that the parts are joined or operate together either directly or indirectly, e.g., through one or more intermediate parts or components, so long as a link occurs. As used herein, “operatively coupled” means that two or more elements are coupled to operate together or are in communication, unidirectional or bidirectional, with one another. As used herein, the term “number” shall mean one or an integer greater than one (i.e., a plurality). As used herein a “set” shall mean one or more.

(36) Although the invention has been described in detail for the purpose of illustration based on what is currently considered to be the most practical and preferred embodiments, it is to be understood that such detail is solely for that purpose and that the invention is not limited to the disclosed embodiments, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present invention contemplates that, to the extent possible, one or more features of any embodiment can be combined with one or more features of any other embodiment.

Claims

1. A method for selecting one or more microphones on a remote speaker/microphone (RSM) assembly for input to a noise reduction algorithm, the method comprising: determining an up orientation for the RSM based on an embedded orientation sensor; accessing a stored fixed mapping of microphone locations, the microphone locations being fixed and known relative to the orientation sensor; determining, for each microphone on the RSM, an up distance relative to the orientation sensor based on the stored fixed mapping of microphone locations and current orientation of the orientation sensor; ranking each microphone on the RSM from the uppermost to the lowermost; selecting the uppermost microphone on the RSM assembly to serve as a voice input microphone; selecting the microphone located furthest from the voice input microphone to serve as a noise input microphone; connecting a first audio input from the selected voice input microphone to a noise reduction algorithm; and connecting a second audio input from the selected noise input microphone to the noise reduction algorithm.
2. The method of claim 1, wherein the orientation sensor includes at least one of an accelerometer or a gyroscope.
3. The method of claim 2, wherein the noise reduction algorithm is an echo cancellation and noise suppression (ECNS) algorithm.
4. The method of claim 2, wherein the noise reduction algorithm is a speech recognition algorithm.
5. The method of claim 1, wherein the RSM assembly includes an even number of microphones organized in pairs, wherein each microphone pair is fixed opposite one another where the first microphone in the pair is furthest from the second microphone in the pair such that any other microphones on the RSM assembly are closer to the either microphone in the pair.
6. The method of claim 1, wherein the internal mapping of microphone locations and orientation sensor is a 3-dimensional database.
7. The method of claim 6, wherein, upon detecting a change in the up orientation for the RSM,

control is returned to the step of determining, for each microphone on the RSM, an up distance relative to the orientation sensor based on the stored fixed mapping of microphone locations and current orientation of the orientation sensor.

8. A system for selecting one or more microphones on a remote speaker/microphone (RSM) assembly for input to a noise reduction algorithm, the system comprising: a set of one or more processors; and a memory operatively coupled to the set of one or more processors and comprising code executable by the set of one or more processors to: determine an up orientation for the RSM based on an embedded orientation sensor; access a stored fixed mapping of microphone locations, the microphone locations being fixed and known relative to the orientation sensor; determine, for each microphone on the RSM, an up distance relative to the orientation sensor based on the stored fixed mapping of microphone locations and current orientation of the orientation sensor; rank each microphone on the RSM from the uppermost to the lowermost; select the uppermost microphone on the RSM assembly to serve as a voice input microphone; select the microphone located furthest from the voice input microphone to serve as a noise input microphone; connect a first audio input from the selected voice input microphone to a noise reduction algorithm; and connect a second audio input from the selected noise input microphone to the noise reduction algorithm.

9. The system of claim 8, wherein the orientation sensor includes at least one of an accelerometer or a gyroscope.

10. The system of claim 9, wherein the noise reduction algorithm is an echo cancellation and noise suppression (ECNS) algorithm.

11. The system of claim 9, wherein the noise reduction algorithm is a Speech Recognition algorithm.

12. The system of claim 8, wherein the RSM assembly includes an even number of microphones organized in pairs, wherein each microphone pair is fixed opposite one another where the first microphone in the pair is furthest from the second microphone in the pair such that any other microphones on the RSM assembly are closer to the either microphone in the pair.

13. The system of claim 8, wherein the internal mapping of microphone locations and orientation sensor is a 3-dimensional database.

14. The system of claim 13, wherein, upon detecting a change in the up orientation for the RSM, control is returned to the executable code that determines, for each microphone on the RSM, an up distance relative to the orientation sensor based on the stored fixed mapping of microphone locations and current orientation of the orientation sensor.

15. A computer program product for selecting one or more microphones on a remote speaker/microphone (RSM) assembly for input to a noise reduction algorithm, comprising: a non-transitory storage medium having computer executable program code comprising: code that determines an up orientation for the RSM based on an embedded orientation sensor; code that accesses a stored fixed mapping of microphone locations, the microphone locations being fixed and known relative to the orientation sensor; code that determines, for each microphone on the RSM, an up distance relative to the orientation sensor based on the stored fixed mapping of microphone locations and current orientation of the orientation sensor; code that ranks each microphone on the RSM from the uppermost to the lowermost; code that selects the uppermost microphone on the RSM assembly to serve as a voice input microphone; code that selects the microphone located furthest from the voice input microphone to serve as a noise input microphone; code that connects a first audio input from the selected voice input microphone to a noise reduction algorithm; and code that connects a second audio input from the selected noise input microphone to the noise reduction algorithm.

16. The computer program product of claim 15, wherein the orientation sensor includes at least one of an accelerometer or a gyroscope.

17. The computer program product of claim 16, wherein the noise reduction algorithm is an echo cancellation and noise suppression (ECNS) algorithm.

18. The computer program product of claim 16, wherein the noise reduction algorithm is a Speech Recognition algorithm.

19. The computer program product of claim 15, wherein the RSM assembly includes an even number of microphones organized in pairs, wherein each microphone pair is fixed opposite one another where the first microphone in the pair is furthest from the second microphone in the pair such that any other microphones on the RSM assembly are closer to the either microphone in the pair.

20. The computer program product of claim 15, wherein the internal mapping of microphone locations and orientation sensor is a 3-dimensional database.

21. The computer program product of claim 20, wherein, upon detecting a change in the up orientation for the RSM, control is returned to the executable code that determines, for each microphone on the RSM, an up distance relative to the orientation sensor based on the stored fixed mapping of microphone locations and current orientation of the orientation sensor.
